An Elementary Proof of the Sobolev Embedding Theorem Without Using Extension Operators

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1 Abstract

When the domain of a function is $\mathbb{R}^N$, the embedding theorem can be established without extension operators, even in the discussion of bounded domains.

2 Introduction

The function $x \mapsto |x|$ belongs to the Sobolev space $W^{1,2}(\Omega)$ on a bounded open interval Ω, which may contain the origin. However, its weak derivatives on the right and left do not join smoothly at the origin. Differentiating weakly once more produces a Dirac delta distribution (up to a constant factor), and the delta distribution is not an L^p function for any $1 \leq p \leq \infty$. For $s \in \mathbb{R}$ with $s < -N/2$, the delta distribution belongs to $W^{s,2}$, but it does not belong to a Sobolev space of nonnegative integer order. Consequently, $x \mapsto |x|$ does not belong to $W^{2,2}(\Omega)$ when the interval contains the origin.

The Fourier transform and inverse Fourier transform turn partial differentiation into multiplication by a polynomial. On the Schwartz space $\mathcal{S}$ of rapidly decreasing smooth functions, whose derivatives remain rapidly decreasing after multiplication by any polynomial, define, for $u \in \mathcal{S}$,

$$\mathcal{F}[u](\xi) := \frac{1}{(2\pi)^{N/2}} \int_{\mathbb{R}^N} e^{-i\xi\cdot x} u(x)\, dx,$$
$$\mathcal{F}^{-1}[u](\xi) := \frac{1}{(2\pi)^{N/2}} \int_{\mathbb{R}^N} e^{i\xi\cdot x} u(x)\, dx.$$

These are continuous linear bijections on $\mathcal{S}$, and integration by parts gives

$$\mathcal{F}[\partial^\alpha u](\xi) = (i\xi)^\alpha \mathcal{F}[u](\xi).$$

The Schwartz space $\mathcal{S} \subset L^2$ is dense in the space L^2 of square-integrable functions, and the extended Fourier transform is an isometry on L^2.

The identity

$$\partial^\alpha u(x) = \mathcal{F}^{-1}\big[(i\xi)^\alpha \mathcal{F}[u](\xi)\big](x)$$

provides motivation for defining derivatives of fractional order.

The inverse Fourier transform really is the inverse of the Fourier transform. In mathematics, before it has been proved that the inverse Fourier transform is the inverse mapping of the Fourier transform, it is sometimes called the conjugate Fourier transform.

3 Main Results

Proposition 1. *Let k, m, N be nonnegative integers satisfying $N \geq 4$ and $m > k + N/2$. Then*

$$W^{m,2}(\mathbb{R}^N) \subset B^k(\mathbb{R}^N),$$

and the embedding is continuous. For every open set $\Omega \subset \mathbb{R}^N$, one has the inclusion

$$W^{m,2}(\Omega) \subset C^k(\Omega).$$

If the first embedding and the second inclusion hold, then the latter embedding is also continuous.

4 Meaning

Sobolev spaces consist of equivalence classes of weakly differentiable functions. Weak smoothness is measured by the integrability of weak derivatives, through the norm used in the definition of a Sobolev space. Even in the weak sense, a function whose regularity is too poor does not belong to the Sobolev space.

If a sufficient degree of weak smoothness is assumed, there exists a representative that is smooth in the ordinary sense, and its partial derivatives are bounded. Roughly speaking, a Sobolev function with sufficient weak smoothness can be made classically smooth by modifying its values on a null set.

The norm on B^k is the sum of the suprema of the absolute values of all partial derivatives of order at most k (or, equivalently, the maximum of these suprema).

Continuity of the embedding means that

$$\|u\|_{B^k} \leq c\|u\|_{W^{m,2}}$$

holds with a finite positive constant c independent of u. If c were allowed to depend on u, such an inequality could always be obtained, and this would cause serious difficulties in the analysis of partial differential equations: an estimate ceases to be useful if the constants multiplying its terms depend on the unknown function.

Since

$$\|u - v\|_{B^k} \leq c\|u - v\|_{W^{m,2}},$$

functions close in the Sobolev topology are also close in the topology of the space of smooth functions. Indeed, for any $\varepsilon > 0$, take $\delta = \varepsilon/(c + 1)$. Then

$$\|u - v\|_{W^{m,2}} < \delta \quad \Longrightarrow \quad \|u - v\|_{B^k} < \varepsilon.$$

Thus the embedding map $u \mapsto u$ from a Sobolev space into B^k or C^k is continuous.

Lemma 1. *There exist constants $C_m, c_m > 0$ such that*

$$c_m(1 + |\xi|^2)^m \leq \sum_{|\alpha| \leq m} |\xi^\alpha|^2 \leq C_m(1 + |\xi|^2)^m.$$

Proof. The upper bound follows from the multinomial theorem. For the lower bound, one may take c_m to be the reciprocal of the largest multinomial coefficient. □

Proof of the proposition. Let $u \in \mathcal{S}$ and put $d_N = \lfloor N/2 \rfloor + 1$. Then $d_N/2 \geq 3/2$. By the Fourier inversion formula and the Cauchy–Schwarz inequality, for $|\alpha| \leq k$,

$$\begin{aligned}|\partial^\alpha u(x)| &\leq c_N \int_{\mathbb{R}^N} (1+|\xi|^2)^{k/2-m/2}(1+|\xi|^2)^{m/2}|\mathcal{F}[u](\xi)|\,d\xi \\ &\leq c_N \int_{\mathbb{R}^N} (1+|\xi|^2)^{-d_N/4}(1+|\xi|^2)^{m/2}|\mathcal{F}[u](\xi)|\,d\xi \\ &\leq c'_N \|u\|_{W^{m,2}}.\end{aligned}$$

Therefore,

$$\|u\|_{B^k} \leq c'_N \|u\|_{W^{m,2}}.$$

Now let $u \in W^{m,2}(\mathbb{R}^N)$. Since $\mathcal{S}$ is dense in $W^{m,2}(\mathbb{R}^N)$, choose $\{u_n\} \subset \mathcal{S}$ converging to u in $W^{m,2}(\mathbb{R}^N)$. By the preceding inequality, $\{u_n\}$ is also a Cauchy sequence in B^k. As B^k is complete, there exists a limit $u' \in B^k$ to which u_n converges uniformly. Uniform convergence implies $u = u'$ almost everywhere. More precisely, for every $\varepsilon > 0$ there is $\nu \in \mathbb{N}$ such that, whenever $n > \nu$,

$$\sup_{x \in \mathbb{R}^N} |u_n(x) - u'(x)| \leq \varepsilon.$$

Thus, for every bounded Jordan-measurable set A,

$$\sup_{x \in A} |u_n(x) - u'(x)| \leq \varepsilon, \qquad \|u_n - u'\|_{L^2(A)} \leq |A|^{1/2}\varepsilon.$$

Since $u_n \to u$ in $L^2(A)$, it follows that $u = u'$ almost everywhere on A. As A is arbitrary, $u = u'$ almost everywhere on $\mathbb{R}^N$. Moreover,

$$\|u_n\|_{B^k} \to \|u'\|_{B^k}, \qquad \|u_n\|_{W^{m,2}} \to \|u\|_{W^{m,2}}.$$

This establishes the embedding inequality.

Take a compactly supported smooth function $\varphi, \psi, \varphi \leq \psi$ that is identically 1 on an open set $\omega \Subset \Omega$. Then $\varphi u \in W^{m,2}(\mathbb{R}^N)$, so $u \in B^k(\omega)$. Since $\|u\|_{B^k(\omega)} \leq \|\varphi u\|_{B^k(\mathbb{R}^N)} \leq c'_N \|\psi u\|_{W^{m,2}(\mathbb{R}^N)}$, $\|u\|_{B^k(\omega)} \leq c'_N \|\psi u\|_{W^{m,2}(\mathbb{R}^N)}$. Next, by taking supremum respect to ψ, $\|u\|_{B^k(\omega)} \leq c'_N \|u\|_{W^{m,2}(\Omega)}$. Since ω can be taken arbitrarily, the asserted inclusion and the continuity of the embedding follow. □

If the boundary of Ω is sufficiently smooth as a manifold (that is, if the boundary has sufficiently good regularity), a bounded linear operator called an *extension operator*

$$E : W^{m,p}(\Omega) \longrightarrow W^{m,p}(\mathbb{R}^N), \qquad (Eu)|_\Omega = u,$$

yields

$$W^{m,2}(\Omega) \subset B^k(\Omega)$$

and continuity of the embedding:

$$\begin{aligned}\|u\|_{B^k(\Omega)} &\leq \|Eu\|_{B^k(\mathbb{R}^N)} \leq c'_N \|Eu\|_{W^{m,2}(\mathbb{R}^N)} \\ &\leq c'_N \|E\|\, \|u\|_{W^{m,2}(\Omega)}.\end{aligned}$$

When the boundary has poor regularity, continuity of the embedding is more difficult to obtain.

For the space

$$W_0^{m,p}(\Omega) := \overline{C_0^\infty(\Omega)}^{W^{m,p}(\Omega)},$$

an embedding theorem into the same space of bounded continuous functions holds without assumptions on boundary smoothness (Miyajima, *Foundations and Applications of Sobolev Spaces*). Under suitable assumptions on the domain Ω and the exponents, products of elements of $W^{m,p}(\Omega)$ are defined almost everywhere in Ω, and $W^{m,p}(\Omega)$ becomes a commutative Banach algebra under a norm equivalent to the Sobolev norm, obtained by multiplying the latter by a suitable constant (Adams–Fournier, *Sobolev Spaces*, Theorem 4.39).

Here,

$$\|E\| = \sup_{\|u\|_{W^{m,2}(\Omega)} \leq 1} \|Eu\|_{W^{m,2}(\mathbb{R}^N)}.$$

A function u is called *Hölder continuous* if there exist $L > 0$ and $0 < \varepsilon_0 < 1$ such that, for every $x, y \in \Omega$,

$$|u(x) - u(y)| \leq L|x - y|^{\varepsilon_0}.$$

When $\varepsilon_0 = 1$, this is called *Lipschitz continuity*. From

$$\frac{|u(x) - u(y)|}{|x - y|} \leq L|x - y|^{\varepsilon_0 - 1},$$

if $\varepsilon_0 > 1$, taking $x \to y$ gives $\nabla u = 0$ wherever the derivative exists; hence u is constant on each connected component. For any $\varepsilon' > 0$, choose

$$\delta = \left(\frac{\varepsilon'}{L}\right)^{1/\varepsilon_0}.$$

Then

$$|x - y| < \delta \quad \Longrightarrow \quad |u(x) - u(y)| \leq \varepsilon'.$$

Thus Hölder continuity implies uniform continuity.

Lipschitz continuity (when differentiable, the absolute values of the partial derivatives are bounded by L, so the slope of the tangent plane cannot change too abruptly) implies uniform continuity (the function varies continuously to the same degree throughout the domain), which implies continuity, which in turn implies sequential continuity. On a compact set, Lipschitz continuity implies Hölder continuity. On an interval of the real line, absolute continuity implies uniform continuity, which implies continuity and hence sequential continuity. Sequential continuity can also be defined in settings more general than metric spaces; this is part of the background to the theory of distributions.

The Hölder space H used in my research consists of functions $u \in B^k(\Omega)$ satisfying

$$\|u\|_H = \sum_{|\alpha| \leq k} \sup_{x \in \Omega} |\partial^\alpha u(x)| + \sup_{\substack{x,y \in \Omega \\ x \neq y}} \frac{|u(x) - u(y)|}{|x - y|^{\varepsilon_0}} < \infty.$$

A Sobolev embedding theorem into a Hölder space also holds under suitable boundary smoothness assumptions. A Hölder space on Ω is a Banach space of functions whose partial derivatives all extend continuously to the boundary.

(References)

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